

Project Phase-I Guidelines - BITE412L - Cloud Computing

Course Instructor: Dr. Priya V

Submission Deadline: 30 July 2026

Abstract

Write a concise abstract (200–300 words) explaining:

Problem statement

Existing challenges

Proposed solution

AWS services used

Literature Survey

Each project must include 15 research papers.

Use papers published in:

- IEEE
- Springer
- Elsevier
- ACM
- Wiley
- MDPI
- Nature
- Scopus Indexed Journals

Prefer papers published between 2023–2026.

Distribution of Papers

Team of 3

Each student studies 5 papers

Student A → Papers 1–5

Student B → Papers 6–10

Student C → Papers 11–15

Team of 2

Student A → Papers 1–8

Student B → Papers 9–15

Individual Project

Single student studies all 15 papers.

Individual Research Gap Analysis

Every student must independently identify:

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- Existing methods
 - Advantages
 - Limitations
 - Research Gap
 - Possible improvement
- Each student should prepare analysis for their assigned five papers.
 - Do NOT copy research gaps directly from the paper.
 - Your analysis should reflect your own understanding.

Suggested Literature Survey Table

Paper	Method	Dataset	Advantages	Limitations	Research Gap
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Project Objectives

Write 4–6 clear objectives.

Example:

Develop an intelligent cloud-based monitoring system.

Reduce response time using Azure services.

Improve prediction accuracy using AI.

Store data securely in the cloud.

Visualize results using dashboards.

Objectives should be measurable and achievable.

Novelty Summary

Students must clearly explain:

What makes your project different from existing work?

Include:

- New feature
- Better algorithm
- Better architecture
- Better AWS integration
- Better security
- Better automation
- Better accuracy

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- Better scalability

Limit to one page.

Proposed Architecture

Two architecture diagrams are mandatory.

Diagram 1- AWS Cloud Architecture

This diagram should explain how AWS services interact.

The architecture should clearly show:

Data flow

Storage

Processing

Authentication

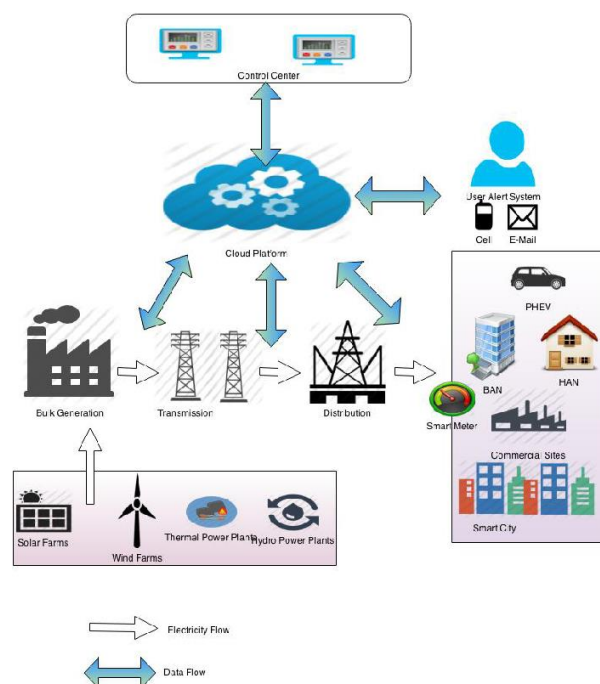
Notifications

Monitoring

Reference - <https://learn.microsoft.com/en-my/azure/architecture/solution-ideas/articles/iot-industrial-solution-architecture>

Diagram 2 - Complete System Architecture

This diagram should explain the overall project workflow.



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Dataset Details

Clearly mention:

- Dataset Name
- Source
- URL
- Size
- Number of Records
- Number of Features
- Data Type
- License
- Purpose of using the dataset
- Preprocessing required

AWS Services Planning

Example: Azure Services Planning for a Healthcare AI Project

Azure Service	Purpose in the Healthcare AI Project
Azure App Service	Host the healthcare web application for doctors and patients
Azure Virtual Machines	Host backend services or specialized healthcare applications
Azure Blob Storage	Store medical images (X-rays, MRI, CT scans), reports, and datasets
Azure SQL Database	Store patient records, appointments, diagnoses, and medical history
Azure Cosmos DB	Store real-time wearable sensor or IoT health monitoring data
Microsoft Entra ID (Azure AD)	Secure authentication and role-based access for doctors, nurses, and patients
Azure API Management	Securely expose APIs to mobile applications and third-party healthcare systems
Azure Functions	Automatically process uploaded medical reports and trigger AI predictions
Azure Machine Learning	Train, evaluate, and deploy disease prediction models
Azure AI Foundry / Azure OpenAI	Provide AI-powered healthcare chatbot, report summarization, and clinical decision support
Azure Data Factory	Collect, clean, and integrate healthcare datasets from multiple sources
Azure Monitor	Monitor application health, performance, and system availability
Azure Service Bus	Process patient requests asynchronously (appointments, lab reports, prescriptions)
Azure Notification Hubs	Send appointment reminders, medication alerts, and emergency notifications
Azure Backup	Automatically back up patient records and databases
Azure Virtual Network (VNet)	Secure communication between Azure services within a private network
Azure Web Application Firewall (WAF)	Protect the healthcare portal from cyberattacks such as SQL Injection and Cross-Site Scripting (XSS)

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Azure Service	Purpose in the Healthcare AI Project
Microsoft Power BI	Visualize patient statistics, disease trends, hospital occupancy, and treatment outcomes

GitHub Guidelines

Each team must create one GitHub repository.

Repository naming format:

ProjectName_Azure_Cloud_Project_2026

Example

SmartFloodMonitoring_Azure_Cloud_Project_2026

Sample GitHub repository structure with task allocation for a team of three students.

Repository Structure

ProjectName_Azure_Project_2026/

```
|
├── README.md
├── docs/
├── literature_survey/
├── architecture/
├── dataset/
├── src/
│   ├── frontend/
│   ├── backend/
│   ├── ai_model/
│   └── azure/
├── results/
├── presentation/
└── references/
```

Git Branch Workflow

```
main
|
develop
├── feature/student1
├── feature/student2
├── feature/student3
└── main
```

- **main** → Final stable version
- **develop** → Integration branch
- **feature/student1** → Student 1 development
- **feature/student2** → Student 2 development
- **feature/student3** → Student 3 development

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Students **must not commit directly to the main branch.**

GitHub Development Workflow

Step 1

Clone the repository.

git clone <repository-url>

Step 2

Switch to your feature branch.

git checkout feature/student1

Step 3

Implement your assigned module.

Step 4

Commit your work regularly.

Example:

git add .

git commit -m "Completed frontend login module"

Step 5

Push changes.

git push origin feature/student1

Step 6

Create a Pull Request (PR)

feature/student1



develop

Step 7

Team members review the Pull Request.

- Review code quality.
- Resolve comments.
- Fix merge conflicts if any.

Step 8

Merge into the **develop** branch.

Step 9

After testing is complete, merge **develop** into **main**.

feature branches



develop



main

Expected Contribution Matrix

Activity	Student 1	Student 2	Student 3
Literature Survey (5 Papers)	✓	✓	✓

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Research Gap (5 Papers)	✓	✓	✓
Frontend Development	✓		
Backend Development		✓	
Database Integration		✓	
Azure Cloud Services	✓	✓	✓
AI / Machine Learning			✓
Testing	✓	✓	✓
Documentation	✓	✓	✓
Presentation	✓	✓	✓
GitHub Commits	✓	✓	✓

Expected GitHub Activity

Each student should have approximately:

- **20–30 meaningful commits**
- **At least 2 Pull Requests**
- **Code review participation**
- **Regular weekly commits**
- **Continuous documentation updates**

Workflow

- Create the repository.
- Create the develop branch.
- Each student creates their own feature/studentX branch.
- Students commit their work regularly to their branch.
- Raise a Pull Request (PR) from the feature branch to develop.
- Team members review the changes.
- Resolve any merge conflicts.
- Merge approved changes into develop.
- Once the project is stable, merge develop into main.
- Tag the release (e.g., v1.0-Phase1).

Important Instruction

- **Hard copy submission is not required.**
- Prepare the complete project report in **soft copy (PDF or Microsoft Word format)**.
- Ensure that all required sections are completed before the review.
- During the project review, each team must **present the soft copy of the document** along with the architecture diagrams, literature survey, research gap analysis, dataset details, GitHub repository, and implementation progress.
- Students should be ready to explain their individual contributions, GitHub commit history, and Azure/AWS service integration during the review.